Phase 2 - Week 5 - Lesson Day 4

Comparing Models + Week 5 Mini Challenge

Today's Goal

This is the last lesson day of Week 5. Today is about comparing models properly, and then putting everything from this week together in one mini challenge.

- Review what Linear and Logistic Regression are each used for
- Learn a simple way to compare two models
- Complete the Week 5 mini challenge

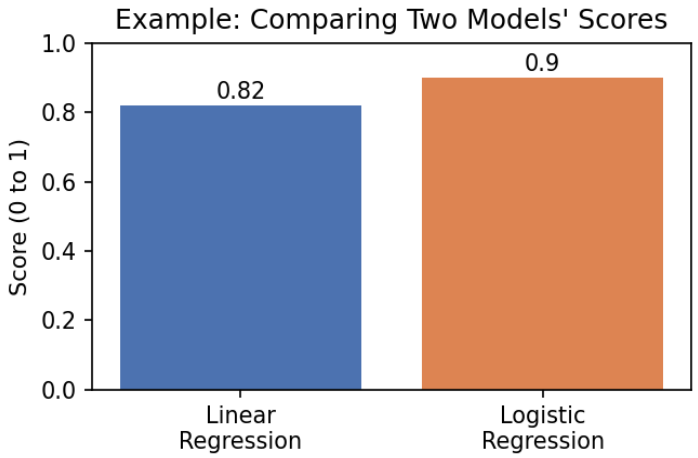
1. Quick Review

Model	Predicts	Example
Linear Regression	A number	Predicting a house price
Logistic Regression	A category (0 or 1)	Predicting pass/fail

Both models use the exact same pattern in code: **.fit()** to train, **.predict()** to make predictions, and **.score()** to check how good the model is.

2. Comparing Two Models

Simple definition: To compare models fairly, you train both of them on the SAME training data, test both on the SAME test data, and compare their **.score()** results. The model with the higher score performed better on this specific data.



Important note: you should only compare models built for the *same kind* of problem. It doesn't make sense to compare a regression model's score directly to a classification model's score, since they measure different things - compare regression models to other regression models, and classification models to other classification models.

```
# Example: comparing two classification models on the SAME data
model_1 = LogisticRegression()
model_1.fit(X_train, y_train)
print("Model 1 score:", model_1.score(X_test, y_test))

model_2 = LogisticRegression(max_iter=500)   # a slightly different setting
model_2.fit(X_train, y_train)
print("Model 2 score:", model_2.score(X_test, y_test))
```

Week 5 Mini Challenge

Open **day-4-comparing-models-mini-challenge.ipynb**. It has a fully solved mini challenge: build one dataset, then train BOTH a Linear Regression model and a Logistic Regression model, each on a version of the problem that fits it.

Mini Challenge Steps

1. Create a dataset of exam scores based on hours studied (with some randomness added).
2. Use the raw scores to train a Linear Regression model that predicts the exact score.
3. Convert the scores into pass/fail (for example, 60 or above = pass) and train a Logistic Regression model on that version.
4. Compare the Linear Regression's score to the Logistic Regression's score, and explain in a comment why they can't be compared directly - they're solving different kinds of problems.
5. Use both trained models to make a prediction for a new, made-up student with a chosen number of hours studied.

Once you and Halimo finish the mini challenge, Week 5 is complete! Next up is Week 6: Tree-Based Models (Decision Trees and Random Forests).